

- Fine-tuning or complex prompting to achieve structured plans

Final Model Selection

After comparative testing, we selected Google Gemini models for production:

- Gemini 1.5 Pro
 - Used for structured 7-day workout generation and plan updating
 - Delivers rich, context-aware, formatted responses reliably
- Gemini Flash
 - Used for generating fast, practical nutrition tips
 - Lightweight and efficient for quick API calls

Reasons for Choosing Gemini:

- API-based, easy to integrate with FastAPI
- Naturally understands structured formatting (days, sections)
- High accuracy with minimal prompt tuning
- Scalable and fast for feedback-based revisions

Activity 1.2. Define the architecture of the application

Architecture Overview: FitBuddy follows a modular FastAPI-based architecture-

- Frontend: HTML + Jinja2 templates for user input and display
- Backend: FastAPI for route handling and logic
- AI Layer: Google Gemini APIs (Pro & Flash)
- Database: SQLite with SQLAlchemy for persistence

Frontend Responsibilities: Users interact via-

- `index.html` – Input form
- `result.html` – Plan, tip, feedback
- `all_users.html` – Admin panel

Backend Responsibilities:

- Receives form data
- Calls Gemini models
- Stores user info and plans
- Renders dynamic templates

AI Integration Points:

- Workout generation & feedback updates → Gemini Pro
- Nutrition tips → Gemini Flash

Activity 1.3 Set up the development Environment:

Install Python & Pip: Ensure Python and pip are installed.

Create Virtual Environment:

```
python -m venv fitbuddy-env  
fitbuddy-env\Scripts\activate (Windows)
```

Install Required Libraries:

```
pip install fastapi uvicorn jinja2 sqlalchemy python-multipart google-generativeai
```

Run Server:

```
uvicorn app.main:app --reload
```

Visit <http://127.0.0.1:8000> for the app and [/docs](#) for API testing.

Set Up Project Structure:

Create folders: [/app](#) (Python files), [/templates](#) (HTML), [/static/images](#) (assets)

```

fitbuddy/
|
├─ requirements.txt      # Project dependencies
|
├─ app/
|   ├─ main.py           # FastAPI entry point
|   ├─ routes.py         # Core route handlers
|   ├─ database.py       # SQLAlchemy models and DB logic
|   ├─ schemas.py        # Pydantic models for validation
|   ├─ gemini_generator.py # Gemini Pro - workout plan generator
|   ├─ gemini_flash_generator.py # Gemini Flash - nutrition tips
|   ├─ updated_plan.py   # Feedback-based plan updater
|   └─ nutrition.py      # Handles nutrition-specific logic (optional)
|
|   └─ templates/
|       ├─ index.html    # User input form
|       ├─ result.html   # Workout plan, tip, feedback
|       └─ all_users.html # Admin dashboard
|
|   └─ static/
|       └─ images/
|           └─ gym-bg.jpg # Gym-themed background
|
└─ fitbuddy.db           # Local SQLite database (auto-generated)

```

Milestone 2. Core functionalities Development

Activity 2.1 Develop the core functionalities:

Workout Plan Generation